Table 2. Development decisions made after observed failures.

Observed problem	**Evidence from testing**	**Change made**	**Reason for regulatory use**
Source text was noisy or poorly structured.	Early app tests and corpus review showed formatting artifacts and document-label problems.	Cleaned source text and verified document identity before indexing.	A regulatory answer is only useful if the cited source is identifiable.
Arbitrary chunks did not match legal structure.	Correct documents were sometimes nearby but answers missed article-level conditions.	Rebuilt the index with article-aware chunks and article metadata.	Users need checkable article citations, not only semantically similar passages.
Dense retrieval alone was weak for exact legal terms.	Terms such as DDCM, RDC numbers, article numbers, and IN 338/2024 required exact matching.	Added BM25 retrieval, reciprocal-rank fusion, and reranking.	Regulatory questions often depend on exact terms and article labels.
English questions over Portuguese documents lost completeness.	Initial English completeness was lower than Portuguese completeness.	Added conservative Portuguese retrieval anchors while preserving answer language.	International users may ask in English while the source corpus remains Portuguese.
The system answered from related but insufficient context.	AI-authored prospective stress testing showed refusal correctness of 0.700.	Tightened the refusal rule and tested a separate refusal-focused AI-authored set.	A related article is not enough when the specific requested fact is absent.
Internal 1.000 values looked too high.	Internal refusal correctness reached 1.000 after targeted development.	Ran an AI-authored prospective stress test and interpreted the internal values as technical checks, not validation proof.	Perfect internal metrics can be a warning sign when benchmark construction, repair, and judging are not fully independent.